

Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

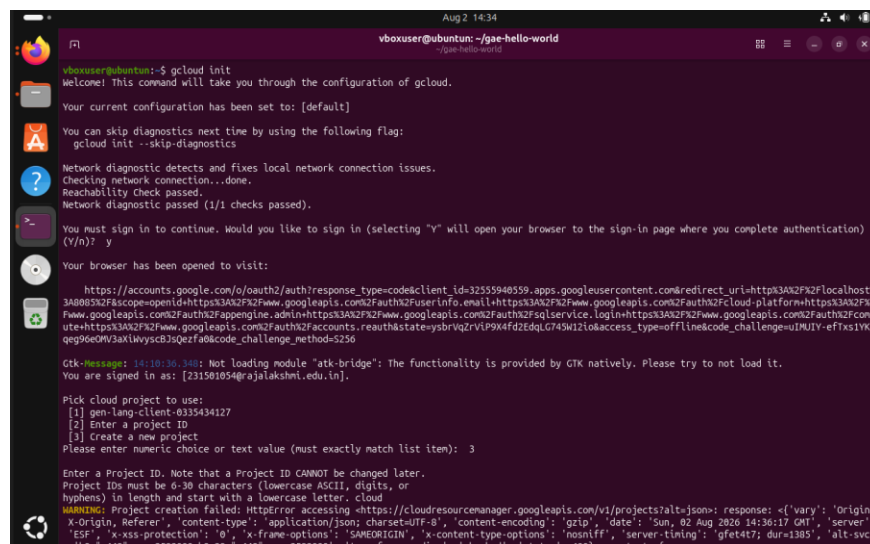
Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-
linux-x86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

`gcloud init`

Sign in to your Google account when prompted.



```
Aug 2 14:34
vboxuser@ubuntu: ~/gae-hello-world
vboxuser@ubuntu:~$ gcloud init
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).

You must sign in to continue. Would you like to sign in (selecting "Y" will open your browser to the sign-in page where you complete authentication)
(Y/N): y

Your browser has been opened to visit:
https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%3A8080%2F&scope=openid%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth%2Fstate%3DysbrVqZrVlP9X4fD2dqlG745K12lo&access_type=offline&code_challenge=UJUIV-efTxs1YKqeg9e0M23ak1WyscB3JQeZfAb&code_challenge_method=S256

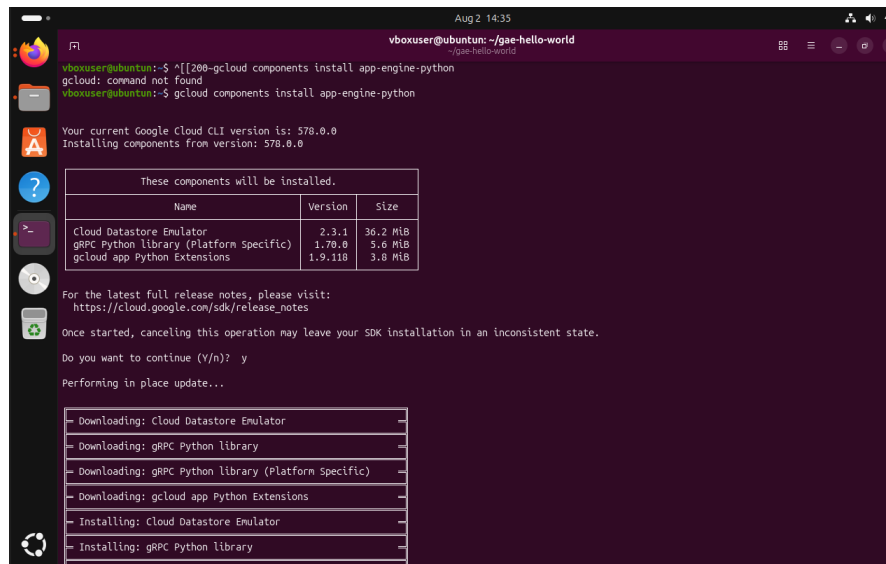
Gtk-Message: 14:10:36.340: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
You are signed in as: [231501054@rajalakshmi.edu.in].

Pick cloud project to use:
[1] gen-lang-client-0335434127
[2] Enter a project ID
[3] create a new project
Please enter numeric choice or text value (must exactly match list item): 3

Enter a Project ID. Note that a Project ID CANNOT be changed later.
Project IDs must be 6-30 characters (lowercase ASCII, digits, or
hyphens) in length and start with a lowercase letter: cloud
WARNING: Project creation failed: HttpError accessing <https://cloudresourcemanager.googleapis.com/v1/projects?alt=json>: response: <{'<var>': 'Origin',
'X-Origin': 'Referer', 'content-type': 'application/json; charset=UTF-8', 'content-encoding': 'gzip', 'date': 'Sun, 02 Aug 2026 14:36:17 GMT', 'server':
'ESF', 'x-xss-protection': '0', 'x-frame-options': 'SAMEORIGIN', 'x-content-type-options': 'nosniff', 'server-timing': 'gfet447; dur=1385', 'alt-svc':
'<var>=447; ma=300000 h3=20; <var>=447; ma=300000'}>, transfer-encoding: 'chunked', 'status': 400}>. content of
```

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gcloud components install app-engine-python



Step 3: Create the App Directory

```
mkdir gae-hello-world  
cd gae-hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route("/")  
def hello():  
    return "Hello World from Google App Engine!"  
  
if __name__ == "__main__":  
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39  
handlers:  
- url: /*  
  script: auto
```

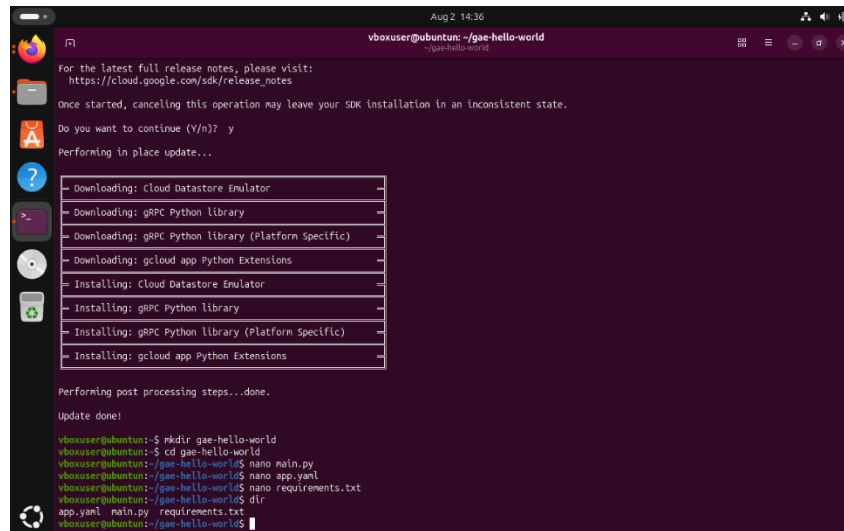
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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The image shows a terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' with a timestamp of 'Aug 2 14:36'. The terminal displays the Google App Engine SDK installation process. It starts with a message about release notes and a confirmation to continue. A progress bar shows the following steps: downloading the Cloud Datastore Emulator, gRPC Python library, gRPC Python library (Platform Specific), and gcloud app Python Extensions; then installing the Cloud Datastore Emulator, gRPC Python library, gRPC Python library (Platform Specific), and gcloud app Python Extensions. After post-processing steps are complete, it says 'Update done!'. The final commands and their outputs are: 'mkdir gae-hello-world', 'cd gae-hello-world', 'nano main.py', 'nano app.yaml', 'nano requirements.txt', and 'dir', all showing successful execution in the '/gae-hello-world' directory.

```
vboxuser@ubuntu: ~/gae-hello-world
Aug 2 14:36

For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, cancelling this operation may leave your SDK installation in an inconsistent state.
Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu: $ mkdir gae-hello-world
vboxuser@ubuntu: $ cd gae-hello-world
vboxuser@ubuntu: ~/gae-hello-world$ nano main.py
vboxuser@ubuntu: ~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu: ~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$ dir
app.yaml  main.py  requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.